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ECON1050 Tutorial 5

Calculus II: Derivatives in Use

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Today's Plan

- ▶ Recap: key ideas from Lecture 5
- ▶ Tutorial questions: Q1–Q5

Recap: Explicit and Implicit Functions

Explicit function

An explicit function writes the dependent variable directly as a function of the independent variable:

$$y = f(x).$$

$y = f(x)$
 $y' = f'(x)$

Implicit function

An implicit relationship is written in the form

$$g(x, y) = c,$$

$g(x, y) = c$
 $xy + x + y = c$ Goal: y'

where y is understood to be a function of x even though it has not been isolated.

Recap: Implicit Differentiation

Procedure

To differentiate an implicitly defined function:

1. Differentiate both sides with respect to x .
2. Treat y as a function of x and apply the chain rule whenever differentiating a term involving y .
3. Use product or quotient rules when required.
4. Solve the resulting equation for $y' = \frac{dy}{dx}$.

$$\begin{aligned}xy + x + y &= C \\1 \cdot y + x \cdot y' + 1 + y' &= 0 \\(x+1)y' &= -1-y \\y' &= \frac{-1-y}{1+x}\end{aligned}$$

Recap: Higher-Order Implicit Derivatives

Second derivative

Once an equation for y' has been obtained, differentiate again with respect to x to find

$$y'' = \frac{d^2y}{dx^2}.$$

Terms containing y or y' must again be differentiated using the chain and product rules as appropriate.

Evaluation at a point

After finding the derivative expression, substitute the specified values of x , y , and, where necessary, y' .

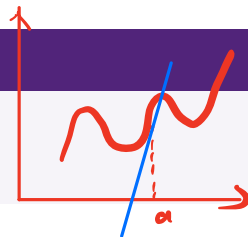
$$\begin{aligned}xy + x + y &= C \\1 \cdot y + x \cdot y' + 1 + y' &= 0 \\y' + y' + x \cdot y'' + 0 + y'' &= 0 \\(1+x)y'' &= -2y' \\y'' &= \frac{-2y'}{1+x} = \frac{-2 \cdot \frac{-1-y}{1+x}}{1+x} = \frac{2+2y}{(1+x)^2}\end{aligned}$$

Recap: Linear Approximation

Linear approximation around $x = a$

For x close to a ,

$$f(x) \approx f(a) + f'(a)(x - a).$$



Interpretation

The linear approximation is the tangent line to f at $x = a$. Therefore, the function and its approximation have the same value and the same first derivative at a .

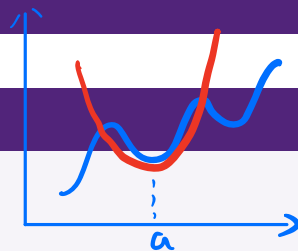
Recap: Quadratic Approximation

Quadratic approximation around $x = a$

If f is twice differentiable,

$$f(x) \approx f(a) + f'(a)(x - a) + \frac{1}{2}f''(a)(x - a)^2.$$

$$\frac{1}{2} \cdot f''(a) \cdot (x - a)^2$$



Matching conditions

At $x = a$, the quadratic approximation matches $f(x)$ in value, first derivative, and second derivative.

Recap: Taylor Polynomial

n th-order approximation around $x = a$

$$f(x) \approx f(a) + \sum_{i=1}^n \frac{f^{(i)}(a)}{i!} (x-a)^i.$$

Equivalently,

$$f(x) \approx f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n.$$

$$n! = n \cdot (n-1) \cdots 1$$

$$\frac{f^{(n)}(a)}{n!} \cdot (x-a)^n$$

Recap: Maclaurin Polynomial and Remainder

Maclaurin polynomial

When the expansion point is $a = 0$, the Taylor polynomial becomes

$$\begin{aligned} f(x) &= f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \cdots + \frac{f^{(n)}(0)}{n!}x^n + R_{n+1}(x) \\ &\approx f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \cdots + \frac{f^{(n)}(0)}{n!}x^n. \end{aligned}$$

$R_{n+1}(x)$

Remainder

Taylor's formula adds a remainder term $R_{n+1}(x)$, which measures the approximation error.

Recap: Elasticity

General definition

For a differentiable function $f(x)$,

$$El_x f(x) = \frac{x f'(x)}{f(x)}.$$

$x \cdot \frac{f'(x)}{f(x)}$

Elasticity measures the percentage responsiveness of $f(x)$ to a percentage change in x .

Demand elasticity

For a demand function $Q(p)$,

$$\varepsilon(p) = \frac{p Q'(p)}{Q(p)}.$$

Recap: Elastic, Unit Elastic and Inelastic

Classification

$ \text{El}_x f(x) > 1$	elastic,
$ \text{El}_x f(x) = 1$	unit elastic,
$ \text{El}_x f(x) < 1$	inelastic,
$ \text{El}_x f(x) = 0$	perfectly inelastic,
$ \text{El}_x f(x) = \infty$	perfectly elastic.

Recap: Elasticity and Revenue

Revenue

If demand is $D(p)$, revenue is

$$R(p) = pD(p).$$

The elasticity of revenue satisfies

$$\text{El}_p R(p) = 1 + \text{El}_p D(p).$$

Implication for price changes

The marginal revenue generated by a price change is positive when demand is inelastic and negative when demand is elastic.

Question 1

5 mins

① Given y is a differentiable function of x , find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ by implicit differentiation when

$$x - y + 3xy = 2.$$

2. Assume that y is a function of x . Find y' for

$$x^3 + y^3 = 4.$$

$$x - y + 3xy = 2$$

$$\textcircled{1} \quad 1 - y' + 3y + 3xy' = 0$$

$$(-1 + 3x) \cdot y' = -1 - 3y$$

$$y' = \frac{-1 - 3y}{-1 + 3x}$$

$$\textcircled{2} \quad 1 - y' + 3y + 3xy' = 0$$

$$0 - y' + 3y' + 3y' + 3xy' = 0$$

$$(-1 + 3x) \cdot y' = -6y'$$

$$y' = \frac{-6y'}{-1 + 3x} = \frac{-6}{-1 + 3x} \cdot \frac{-1 - 3y}{-1 + 3x} \\ = \frac{6 + 18y}{(-1 + 3x)^2}$$

Tutorial Q1.1: First Derivative

Knowledge used

- ▶ Implicit differentiation: differentiate both sides with respect to x .
- ▶ Product rule for the term xy .
- ▶ Treat y as a function of x , so $\frac{dy}{dx} = y'$.

Solution

From

$$x - y + 3xy = 2,$$

differentiate with respect to x :

$$1 - y' + 3y + 3xy' = 0.$$

Collect the y' terms:

$$y'(-1 + 3x) = -1 - 3y.$$

Therefore,

$$y' = \frac{1 + 3y}{1 - 3x}.$$

Tutorial Q1.1: Second Derivative

Knowledge used

- ▶ Differentiate the first-derivative equation again.
- ▶ Product rule for xy' .
- ▶ Substitute the previously obtained expression for y' .

Solution

Differentiate

$$1 - y' + 3y + 3xy' = 0$$

again:

$$-y'' + 3y' + 3y' + 3xy'' = 0.$$

Hence

$$y''(3x - 1) = -6y' \implies y'' = \frac{6y'}{1 - 3x}.$$

Substitute $y' = \frac{1+3y}{1-3x}$:

$$y'' = \frac{6 + 18y}{(1 - 3x)^2}.$$

Tutorial Q1.2: Implicit Differentiation

Knowledge used

- ▶ Chain rule for a power of an implicitly defined variable.
- ▶ $\frac{d}{dx}(y^3) = 3y^2 y'$.

Solution

From

$$x^3 + y^3 = 4,$$

differentiate with respect to x :

$$3x^2 + 3y^2 y' = 0.$$

Solve for y' :

$$3y^2 y' = -3x^2 \quad \Rightarrow \quad \boxed{y' = -\frac{x^2}{y^2}}.$$

Tutorial Q2

Question 2

3 min 30 s

Assume y is a differentiable function of x that satisfies

$$2x^2 + 6xy + y^2 = 18.$$

Find y' and y'' at the point

$$(x, y) = (1, 2).$$

$$2x^2 + 6xy + y^2 = 18$$

$$\textcircled{1} \quad 4x + 6y + 6xy' + 2y \cdot y' = 0$$

$$2x + 3y + 3xy' + y \cdot y' = 0$$

$$(3x + y) \cdot y' = -2x - 3y$$

$$y' = \frac{-2x - 3y}{3x + y}$$

$$y'(1, 2) = \frac{-2 - 6}{3 + 2} = \frac{-8}{5}$$

$$\textcircled{2} \quad 2x + 3y + 3xy' + y \cdot y' = 0$$

$$2 + 3y' + 3y' + 3xy'' + y' \cdot y' + y \cdot y'' = 0$$

$$(3x + y) \cdot y'' = -2 - 6y' - (y')^2$$

$$y'' = \frac{-2 - 6y' - (y')^2}{3x + y}$$

$$y''(1, 2) = \frac{-2 - 6 \cdot (-\frac{8}{5}) - (-\frac{8}{5})^2}{3 + 2}$$

Tutorial Q2: First Derivative

Knowledge used

- ▶ Implicit differentiation.
- ▶ Product rule for xy .
- ▶ Chain rule for y^2 .

Solution

Differentiate:

$$4x + 6y + 6xy' + 2yy' = 0.$$

Thus

$$y'(6x + 2y) = -4x - 6y,$$

so

$$y' = -\frac{2x + 3y}{3x + y}.$$

At $(1, 2)$,

$$y'(1, 2) = -\frac{8}{5}.$$

Tutorial Q2: Second Derivative

Knowledge used

- ▶ Quotient rule applied to the expression for y' .
- ▶ Evaluate the resulting expression at $(x, y) = (1, 2)$.

Solution

Starting from

$$y' = -\frac{2x + 3y}{3x + y},$$

the tutorial solution simplifies the second derivative to

$$y'' = \frac{42xy + 7y^2 + 14x^2}{(3x + y)^3}.$$

Therefore,

$$y''(1, 2) = \frac{42(1)(2) + 7(2)^2 + 14(1)^2}{(3 + 2)^3} = \boxed{\frac{126}{125}}.$$

Tutorial Q3

Question 3

5 mins

Find the linear and quadratic approximations to

$$f(x) = (1+x)^5$$

Linear: $f(x) \approx f(a) + f'(a) \cdot (x-a)$
Quadratic: $f(x) \approx f(a) + f'(a)(x-a) + \frac{f''(a)}{2}(x-a)^2$

about $x = 0$, and use them to approximate

$$1.05^5.$$

$$f(x) = (1+x)^5 \Rightarrow f(0) = 1$$

$$f'(x) = 1.5 \cdot (1+x)^4 \Rightarrow f'(0) = 5$$

$$f''(x) = 5 \cdot 1.4 \cdot (1+x)^3 = 20(1+x)^3 \Rightarrow f''(0) = 20$$

$$1.05^5 = (1+0.05)^5 = f(0.05)$$

$$\text{Linear: } f(x) \approx 1 + 5 \cdot x$$

$$\Rightarrow f(0.05) \approx 1 + 5 \cdot 0.05 = 1.25$$

$$\text{Quadratic: } f(x) \approx 1 + 5 \cdot x + 10 \cdot x^2$$

$$\Rightarrow f(0.05) \approx 1 + 5 \cdot 0.05 + 10 \cdot (0.05)^2 = 1.275$$

Tutorial Q3: Linear Approximation

Knowledge used

- ▶ Linear approximation:

$$f(x) \approx f(a) + f'(a)(x - a).$$

- ▶ Here the expansion point is $a = 0$.

Solution

For

$$f(x) = (1 + x)^5, \quad f'(x) = 5(1 + x)^4,$$

we have

$$f(0) = 1, \quad f'(0) = 5.$$

Hence

$$f(x) \approx 1 + 5x.$$

With $x = 0.05$,

$$1.05^5 \approx 1.25.$$

Tutorial Q3: Quadratic Approximation

Knowledge used

- Quadratic approximation:

$$f(x) \approx f(a) + f'(a)(x - a) + \frac{1}{2}f''(a)(x - a)^2.$$

- Compare the linear and quadratic approximations at the same nearby value.

Solution

Since

$$f'(x) = 20(1 + x)^3, \quad f'(0) = 20,$$

we obtain

$$f(x) \approx 1 + 5x + 10x^2.$$

At $x = 0.05$,

$$1.05^5 \approx 1 + 5(0.05) + 10(0.05)^2 = 1.275.$$

The calculator value is

$$1.05^5 = 1.276281563.$$

Tutorial Q4

Question 4

5 mins

Let

$$f(x) = \sqrt{2x}.$$

Find the third-order Taylor polynomial of $f(x)$ about

$$f(x) \approx f(a) + f'(a)(x-a) + \frac{f''(a)}{2}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3$$

$$a = 1.$$

$$f(x) = \sqrt{2x} = \sqrt{2} \cdot x^{\frac{1}{2}} \Rightarrow f(1) = \sqrt{2}$$

$$f'(x) = \frac{\sqrt{2}}{2} \cdot x^{-\frac{1}{2}} \Rightarrow f'(1) = \frac{\sqrt{2}}{2}$$

$$f''(x) = \frac{\sqrt{2}}{2} \cdot (-\frac{1}{2}) \cdot x^{-\frac{3}{2}} = -\frac{\sqrt{2}}{4} \cdot x^{-\frac{3}{2}} \Rightarrow f''(1) = -\frac{\sqrt{2}}{4}$$

$$f'''(x) = -\frac{\sqrt{2}}{4} \cdot (-\frac{3}{2}) \cdot x^{-\frac{5}{2}} = \frac{3\sqrt{2}}{8} \cdot x^{-\frac{5}{2}} \Rightarrow f'''(1) = \frac{3\sqrt{2}}{8}$$

$$\begin{aligned} f(x) &\approx \sqrt{2} + \frac{\sqrt{2}}{2} \cdot (x-1) + (-\frac{\sqrt{2}}{4}) \cdot \frac{1}{2} \cdot (x-1)^2 + \frac{3\sqrt{2}}{8} \cdot \frac{1}{3 \cdot 2} \cdot (x-1)^3 \\ &= \sqrt{2} + \frac{\sqrt{2}}{2} (x-1) - \frac{\sqrt{2}}{8} (x-1)^2 + \frac{\sqrt{2}}{16} (x-1)^3 \end{aligned}$$

Tutorial Q4: Derivatives at the Expansion Point

Knowledge used

- ▶ Third-order Taylor polynomial around a .
- ▶ Power rule and chain rule to obtain f' , f'' , and f''' .

Solution

Write

$$f(x) = (2x)^{1/2}.$$

Then

$$f'(x) = (2x)^{-1/2},$$

$$f''(x) = -(2x)^{-3/2},$$

$$f'''(x) = 3(2x)^{-5/2}.$$

At $a = 1$,

$$f(1) = \sqrt{2}, \quad f'(1) = \frac{1}{\sqrt{2}}, \quad f''(1) = -\frac{1}{\sqrt{2^3}}, \quad f'''(1) = \frac{3}{\sqrt{2^5}}.$$

Tutorial Q4: Third-Order Taylor Polynomial

Knowledge used

- Substitute derivative values into

$$f(x) \approx f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3.$$

Solution

For $a = 1$,

$$\sqrt{2x} \approx \sqrt{2} + \frac{1}{\sqrt{2}}(x-1) - \frac{1}{2\sqrt{2^3}}(x-1)^2 + \frac{1}{2\sqrt{2^5}}(x-1)^3.$$

Question 5

7 mins

For each demand function, calculate elasticity as a function of price and find the conditions under which demand is elastic.

1. $Q(p) = a - bp$, where $a > 0$ and $b > 0$.

2. $Q(p) = p^{-\lambda}$, where $\lambda > 0$.

3. $Q(p) = 100 - 4p^2$.

$$\text{elastic} \begin{cases} |\epsilon| > 1: \text{elastic} \\ |\epsilon| = 1: \text{unit elastic} \end{cases}$$

$$|\epsilon(p)| \geq 1$$

Use

$$\epsilon(p) = \frac{pQ'(p)}{Q(p)}.$$

$$(a) \quad Q(p) = a - bp$$

$$Q'(p) = -b$$

$$\epsilon(p) = \frac{p \cdot (-b)}{a - bp} = \frac{-bp}{a - bp}$$

$$|\epsilon(p)| = \left| \frac{-bp}{a - bp} \right| = \frac{bp}{a - bp} \geq 1$$

$$bp \geq a - bp$$

$$2bp \geq a$$

$$p \geq \frac{a}{2b}$$

$$(h) Q(p) = p^{-\lambda}, \lambda > 0$$

$$Q'(p) = -\lambda \cdot p^{-\lambda-1}$$

$$E(p) = \frac{p \cdot (-\lambda) \cdot p^{-\lambda-1}}{p^{-\lambda}} = -\lambda \cdot p^{1-\lambda-1+\lambda} = -\lambda$$

$$|E(p)| = |-\lambda| = \underline{\lambda \geq 1}$$

$$(c) Q(p) = 100 - 4p^2$$

$$Q'(p) = -8p$$

$$E(p) = \frac{p \cdot (-8p)}{100 - 4p^2} = \frac{-8p^2}{100 - 4p^2}$$

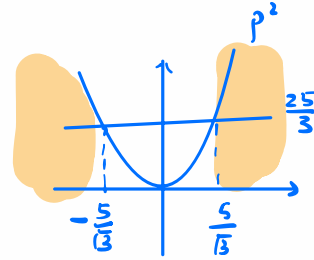
$$|E(p)| = \left| \frac{-8p^2}{100 - 4p^2} \right| = \frac{8p^2}{100 - 4p^2} \geq 1$$

$$8p^2 \geq 100 - 4p^2$$

$$12p^2 \geq 100$$

$$p^2 \geq \frac{25}{3}$$

$$\underline{p \in \mathbb{R} \setminus (-\frac{5}{\sqrt{3}}, \frac{5}{\sqrt{3}})}$$



Tutorial Q5(a): Linear Demand

Knowledge used

- ▶ Elasticity: $\varepsilon(p) = \frac{pQ'(p)}{Q(p)}$.
- ▶ Following the tutorial solution, demand is elastic when $|\varepsilon(p)| \geq 1$.

Solution

For $Q(p) = a - bp$, $Q'(p) = -b$, so

$$\varepsilon(p) = -\frac{bp}{a - bp}.$$

Then

$$\left| -\frac{bp}{a - bp} \right| \geq 1 \implies bp \geq a - bp \implies p \geq \frac{a}{2b}.$$

Tutorial Q5(b): Constant Elasticity Demand

Knowledge used

- ▶ Apply the power rule to $Q(p) = p^{-\lambda}$.
- ▶ Substitute the derivative into the elasticity formula.

Solution

For

$$Q(p) = p^{-\lambda},$$

$$Q'(p) = -\lambda p^{-\lambda-1}.$$

Therefore,

$$\varepsilon(p) = \frac{p(-\lambda p^{-\lambda-1})}{p^{-\lambda}} = \boxed{-\lambda}.$$

Thus, following the tutorial solution, demand is elastic for all p if

$$\boxed{\lambda \geq 1}.$$

Tutorial Q5(c): Quadratic Demand

Knowledge used

- ▶ Use the power rule and the elasticity formula.
- ▶ Following the tutorial solution, solve $|\varepsilon(p)| \geq 1$.

Solution

For $Q(p) = 100 - 4p^2$, $Q'(p) = -8p$, hence

$$\varepsilon(p) = -\frac{8p^2}{100 - 4p^2}.$$

The elasticity condition gives

$$\left| -\frac{8p^2}{100 - 4p^2} \right| \geq 1 \implies 8p^2 \geq 100 - 4p^2 \implies p^2 \geq \frac{100}{12}.$$

Therefore,

$$p \geq \frac{5}{\sqrt{3}}.$$

Key Takeaways

- ▶ Implicit differentiation treats y as a function of x and uses the chain rule whenever y appears.
- ▶ Linear, quadratic and Taylor polynomials approximate a function locally using its derivatives at the expansion point.
- ▶ Elasticity measures percentage responsiveness and is computed as $\frac{xf'(x)}{f(x)}$.
- ▶ For demand, elasticity helps classify how strongly quantity responds to price changes.